

Weekly Progress Report - Week 03

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Domain: Machine Learning
Project Name: Smart City Traffic Forecasting
Week Ending: 03
Submission Date: 11-06-26

I. Overview

During Week 3, the Smart City Traffic Forecasting project focused on improving model performance through hyperparameter tuning, analyzing feature importance, and evaluating prediction accuracy. The objective was to enhance forecasting reliability and better understand the factors influencing traffic flow.

II. Achievements

- 1. Model Optimization
 - Performed hyperparameter tuning using GridSearchCV.
 - Optimized Random Forest model parameters for improved forecasting accuracy.
- 2. Feature Engineering and Analysis
 - Evaluated the contribution of key features such as Junction, Hour, Day, Month, and DayOfWeek.
 - Generated feature importance visualizations for model interpretation.
- 3. Prediction Analysis
 - Compared actual traffic values with predicted traffic values.
 - Generated prediction visualizations to assess forecasting performance.
- 4. Model Improvement
 - Improved model robustness and forecasting capability.
 - Saved the optimized Random Forest model for final deployment and testing.

Figure-1: Feature Importance Analysis

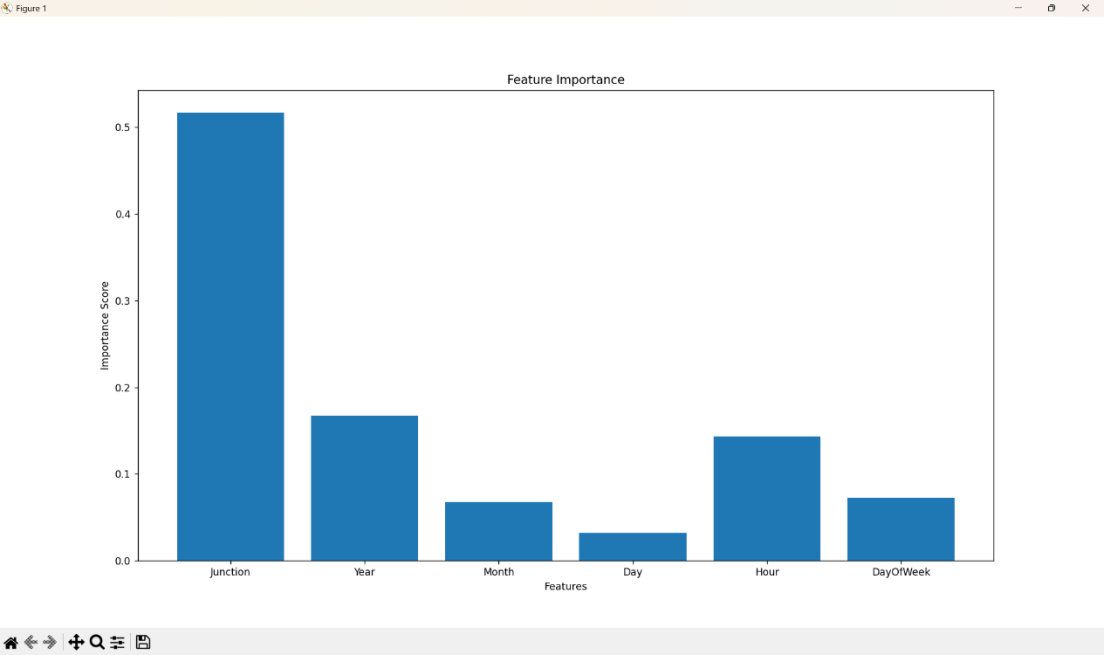


Figure-2 Actual vs Predicted Traffic

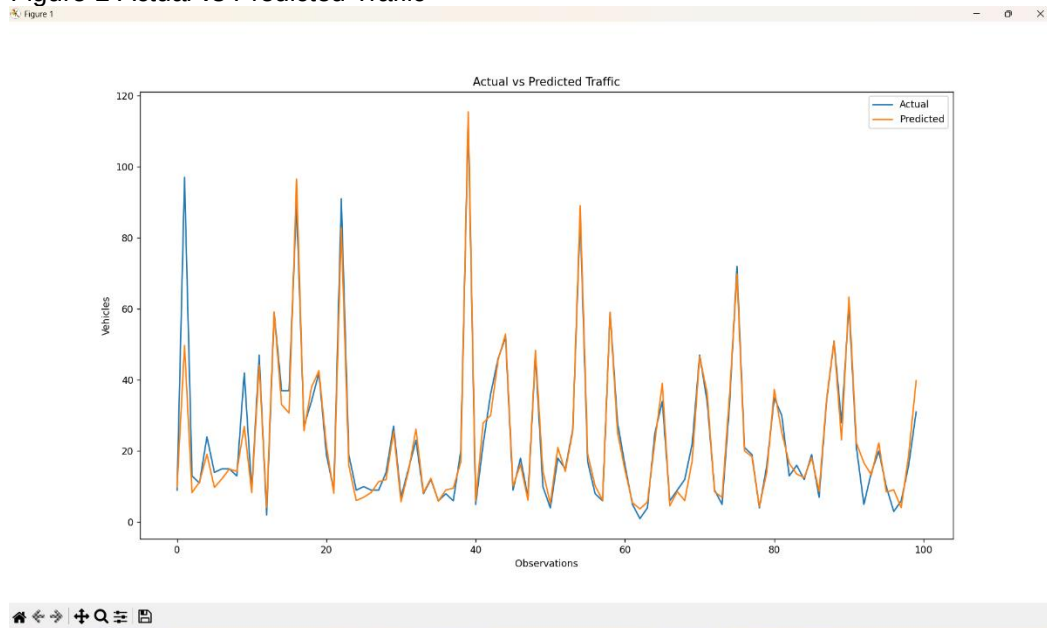
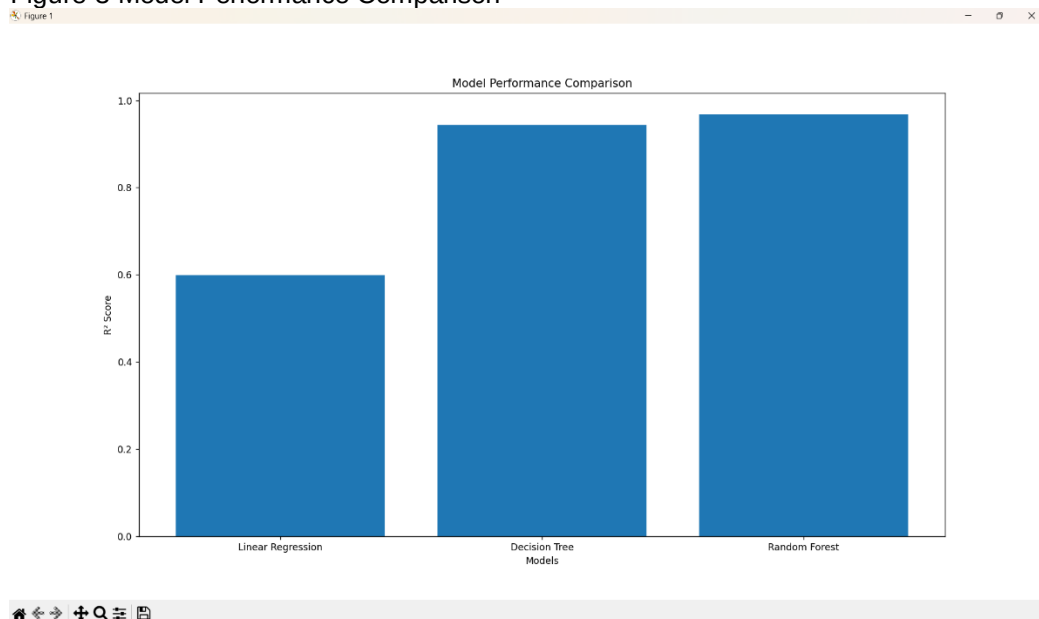


Figure-3 Model Performance Comparison



III. Challenges

- Selecting suitable hyperparameters for model optimization.
- Balancing model accuracy and training time.
- Interpreting feature importance and prediction trends effectively.

IV. Learning Resources

- Scikit-Learn GridSearchCV Documentation
- Random Forest Regressor Documentation
- Feature Importance Analysis Resources
- Machine Learning Model Evaluation Tutorials

V. Next Week Goals

- Finalize the forecasting model.
- Test the model on unseen data.

- Prepare the final project report and presentation.
- Organize GitHub repository and project documentation.
- Complete internship project submission requirements.

VI. Additional Comments

Week 3 strengthened the forecasting pipeline through model optimization and detailed performance analysis. The project is progressing toward final deployment and comprehensive evaluation.